Generative Modeling with Normalizing Flows

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Outline

- Optimal Transport Setup
- Mesh-Free Flow Construction (Tabak–Turner style)
- Solving for β at Each Step
- Choosing the Basis F_ℓ^k and Bandwidth α
- Constructing the Transport Map
- Baseline Results (RBF Flow)
- Two Adaptations & Two Approaches
- Conclusion

Monge's Optimal Transport Problem

Goal: Find a transport map $y = y(x) : \mathbb{R}^n \to \mathbb{R}^n$ that pushes a source density $\rho(x)$ to a target density $\mu(y)$, minimizing the total transportation cost.

Cost function:

$$M(y) = \int_{\mathbb{R}^n} c(x, y(x))\rho(x)\,dx$$

Change of variables:

$$\rho(x) = J^y(x)\mu(y(x)), \quad \text{where } J^y(x) = \det(\nabla y(x))$$

Definition: A map $y(x)$ that satisfies the above condition and minimizes $M(y)$ is called an *optimal map*.

Brenier's Theorem

Let ρ and μ be absolutely continuous probability densities on $\mathbb{R}^d$ with finite second moments.

Brenier's Theorem: In the case of the *quadratic cost*

$$c(x, y) = \frac{1}{2}\|x - y\|^2,$$

there exists a **unique** optimal transport map

$$T^* : \mathbb{R}^d \to \mathbb{R}^d$$

that pushes ρ forward to μ.

Moreover, this optimal map is given by the gradient of a convex potential:

$$T^*(z) = \nabla\phi(z),$$

where $\phi : \mathbb{R}^d \to \mathbb{R}$ is a convex function.

Mesh-Free Implementation

In the case of the quadratic cost

$$c(x, y) = \frac{1}{2}\|x - y\|^2,$$

Brenier's theorem implies that the optimal transport map is given by the gradient of a convex potential:

$$T(x) = \nabla U(x),$$

where U is a convex function, often called the *Brenier potential*.

We construct this transport map iteratively. Let $z^k(x)$ denote the map at iteration k, initialized by

$$z^0(x) = x.$$

At each step, the map is updated using the current transport, producing a sequence of deformations of the source samples.

As the flow evolves, these maps converge to the optimal transport:

$$\lim_{k\to\infty} z^k(x) = y(x).$$

Mesh-Free Implementation

We consider the case where the densities $\rho(x)$ and $\mu(y)$ are only known through samples:

$$x_i \sim \rho(x), \quad y_j \sim \mu(y), \quad i = 1, \ldots, m, \quad j = 1, \ldots, n$$

Starting from these samples, we construct a sequence of transport maps $z^k(x)$ that progressively move the source samples toward the target distribution, without requiring a mesh or density evaluation.

Instead of following the full gradient flow, we restrict the maps at each step to a parametrized form:

$$z_i^{k+1} = \nabla U_\beta^k(z_i^k).$$

Mesh-Free Implementation

The potential is:

$$U_\beta^k(z) = \frac{1}{2}\|z\|^2 + \sum_l \beta_l F_l^k(z).$$

Therefore, the gradient of the potential is:

$$\nabla U_\beta^k(z) = z + \sum_l \beta_l \nabla F_l^k(z).$$

The quadratic term $\frac{1}{2}\|z\|^2$ is strongly convex, and each basis function F_l^k is chosen to be convex. Convexity is preserved under nonnegative linear combinations, so the potential U_β^k is convex when $\beta_l \geq 0$.

Computing β: Dual Objective

At iteration k, we restrict the transport map to the form

$$z_i^{k+1} = \nabla U_\beta^k(z_i^k),$$

where the potential is parameterized by coefficients $\beta = (\beta_l)$.

The dual objective is defined as

$$\hat{D}^k(\beta) = \frac{1}{m}\sum_{i=1}^{m} U_\beta^k(z_i^k) \; + \; \frac{1}{n}\sum_{j=1}^{n} (U_\beta^k)^*(y_j),$$

where $(U_\beta^k)^*$ denotes the convex conjugate of U_β^k.

Maximizing $\hat{D}^k(\beta)$ selects the coefficients β that best align the transported source samples z_i^k with the target samples y_j at the current iteration.

This optimization is finite-dimensional and convex, since it is performed over the coefficients β_l for a fixed set of particles z_i^k.

Computing β via Gradient Ascent

To maximize the dual objective $\hat{D}^k(\beta)$, we compute the gradient with respect to each coefficient β_l:

$$\frac{\partial \hat{D}^k(\beta)}{\partial \beta_l} = \frac{1}{m}\sum_{i=1}^{m} F_l^k(z_i^k) - \frac{1}{n}\sum_{j=1}^{n} F_l^k(y_j).$$

We then apply a single step of gradient ascent:

$$\bar{\beta}_l \leftarrow \bar{\beta}_l + \eta \cdot \frac{\partial \hat{D}^k(\beta)}{\partial \beta_l}.$$

Substituting the expression for the gradient gives

$$\bar{\beta}_l \leftarrow \bar{\beta}_l + \eta \left(\frac{1}{m}\sum_{i=1}^{m} F_l^k(z_i^k) - \frac{1}{n}\sum_{j=1}^{n} F_l^k(y_j) \right).$$

At convergence (as $k \to \infty$), the residual vanishes:

$$\frac{1}{m}\sum_{i=1}^{m} F_l^\infty(z_i^\infty) = \frac{1}{n}\sum_{j=1}^{n} F_l^\infty(y_j).$$

Intuition

We parameterize the Brenier potential as

$$U_\beta^k(z) = \tfrac{1}{2}\|z\|^2 + \sum_l \beta_l F_l^k(z),$$

where each F_l^k is a **convex perturbation**. The gradient of the dual objective with respect to β_l is

$$\frac{\partial \hat{D}^k(\beta)}{\partial \beta_l} = \frac{1}{m}\sum_i F_l^k(z_i^k) - \frac{1}{n}\sum_j F_l^k(y_j).$$

This quantity measures whether the current potential causes the **transported source mass** to occupy regions emphasized by F_l^k more or less than the **target mass** does.

Gradient descent adjusts β_l to remove this mismatch. When the gradient vanishes, the Brenier potential is correctly shaped **along that convex deformation direction**. Enforcing this condition for all l aligns the overall shape of the potential.

Choosing Basis Functions $F_l(z)$

To complete the procedure, we must specify the form and number of functions $F_l(z)$ used at each time step. Key considerations:

- Avoid over- or under-resolving the densities $\rho^k(z)$ and $\mu(y)$.
- Functions F_l should perform well in high-dimensional settings.

Example: A radially symmetric convex perturbation,

$$F_l(z) = r \,\mathrm{erf}\left(\frac{r}{\alpha}\right) + \frac{\alpha}{\sqrt{\pi}} e^{-(r/\alpha)^2}, \qquad r = \|z - \tilde{z}_l\|.$$

This choice satisfies the key considerations:

- Radial symmetry avoids coordinate bias and scales naturally in high dimensions.
- Convexity ensures the Brenier potential remains convex.
- The perturbation width α expands in low-density regions to avoid reacting to noise and contracts in high-density regions to capture fine structure, preventing both over- and under-resolution.

Choosing the Bandwidth α

The bandwidth α is chosen adaptively based on local densities:

$$\alpha \propto \left(\frac{n_p}{n+m} \left(\frac{1}{\rho(\tilde{z}_l)} + \frac{1}{\mu(\tilde{z}_l)} \right) \right)^{1/d}.$$

Intuition:

- In low-density regions, $\rho(\tilde{z}_l)$ or $\mu(\tilde{z}_l)$ is small, so α becomes large, producing a wider perturbation.
- This allows mass to move farther when data are sparse and large corrections are needed.
- In high-density regions, α shrinks, preventing overcorrection and preserving fine structure.

The factor n_p controls how many particles each perturbation influences, setting the locality of the update and balancing resolution against stability.

Constructing the Transport Map with the RBF

We restrict the transport map to a parameterized potential:

$$U_\beta^k(z) = \frac{1}{2}\|z\|^2 + \sum_l \beta_l F_l^k(z)$$

Its gradient defines the update step:

$$\nabla U_\beta^k(z) = z + \sum_l \beta_l \nabla F_l^k(z)$$

$$z_i^{k+1} = \nabla U_\beta^k(z_i^k)$$

Using a radially symmetric basis

$$F_l(z) = r\,\mathrm{erf}(r/\alpha) + \frac{\alpha}{\sqrt{\pi}} e^{-(r/\alpha)^2},$$

we obtain the explicit update:

$$z^{k+1} = z^k + \sum_l \beta_l f_l(r)(z^k - \tilde{z}_l)$$

where

$$f_l(r) = \frac{1}{r}\frac{dF}{dr} = \frac{\mathrm{erf}(r/\alpha)}{r}, \quad r = \|z^k - \tilde{z}_l\|.$$

Gradient as Sum of Convex Functions

The transport map is defined as

$$\nabla U_{\beta}^{k}(z) = z + \sum_{l} \beta_l \nabla F_l^k(z),$$

which is the gradient of the potential

$$U_{\beta}^{k}(z) = \frac{1}{2}\|z\|^2 + \sum_{l} \beta_l F_l^k(z).$$

This can be reformulated as

$$\nabla U_{\beta}^{k}(z) = \nabla \left(\frac{1}{2}\|z\|^2 + \sum_{l} \beta_l F_l^k(z) \right).$$

Iterative Viewpoint of the Transport Map

We start from the identity map

$$z^0(x) = x.$$

We then apply successive transport updates:

$$z^1(x) = \nabla U_\beta^0(z^0(x)) = x + \mathrm{pert}_0,$$

$$z^2(x) = \nabla U_\beta^1(z^1(x)) = z^1(x) + \mathrm{pert}_1,$$

$$z^3(x) = \nabla U_\beta^2(z^2(x)) = z^2(x) + \mathrm{pert}_2.$$

After k steps, the transported point can be written as

$$z^k(x) = x + \mathrm{pert}_0 + \mathrm{pert}_1 + \cdots + \mathrm{pert}_{k-1}.$$

In each step, the center $\tilde{z}_l$ of the basis function F_l^k is chosen randomly (either from the current particles or from the target distribution) to localize the transport update.

Normalizing Flow Results

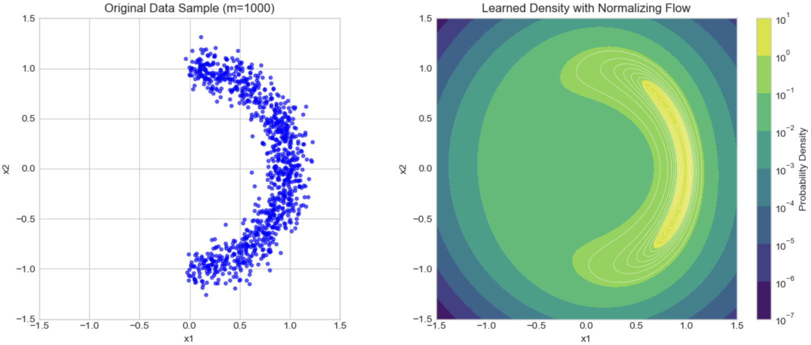

Figure: Results of the Normalizing flow after 1000 steps on a crescent distribution

$$\alpha = \sqrt{2\pi}\left(\Omega_n^{-1}\frac{n_p}{m}\right)^{\frac{1}{n}} e^{\frac{||x_0||^2}{2n}}, \quad f(r) = \frac{1}{\alpha}\frac{\operatorname{erf}\left(\frac{r}{\alpha}\right)}{\frac{r}{\alpha}}.$$

Normalizing Flow Results

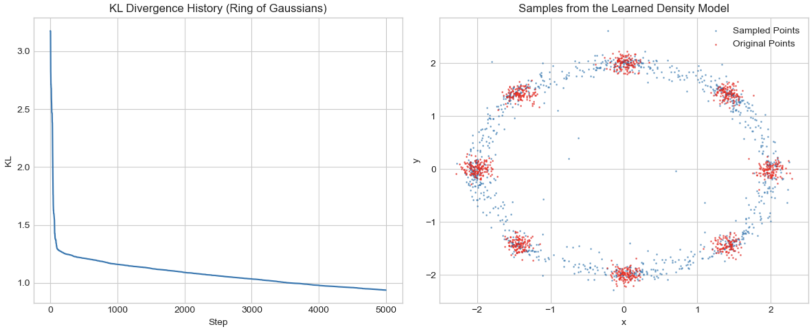

Figure: Results of the Normalizing flow after 5000 steps on a ring distribution

Adaptations to the Tabak–Turner Model

We modify the Tabak–Turner flow in two ways.

First, we take the gradient with respect to the initial point rather than the transported point:

$$z_i^{k+1} = z_i^k + \beta_k \nabla_{z^0} F_\ell^k \left(\| z_i^0 - \tilde{z}^k \| \right).$$

Second, we enforce convexity throughout the flow using linear programming. These two modifications together constitute our first approach.

Our second approach replaces the basis functions F_ℓ with Input Convex Neural Networks (ICNNs).

Checking Convexity for Adaptive β_l

After adding a new basis function, the updated potential is

$$U_\beta^{k+1}(z) = U_\beta^k(z) + \beta_{k+1} F_{k+1}(z),$$

where U_β^k and F_{k+1} are convex. If $\beta_{k+1} < 0$, convexity of U_β^{k+1} is no longer guaranteed. Rather than checking convexity everywhere, we test it empirically on the fixed initial sample locations:

$$\{(z_0^{(i)}, U_\beta^{k+1}(z_0^{(i)}))\}_{i=1}^m.$$

These points come from a convex function if and only if, for each i, there exists a subgradient $g_i \in \mathbb{R}^d$ such that

$$U_\beta^{k+1}(z_0^{(j)}) \;\geq\; U_\beta^{k+1}(z_0^{(i)}) + g_i^\top (z_0^{(j)} - z_0^{(i)}) \quad \forall j.$$

If this condition fails, we reduce the magnitude of β_{k+1} (e.g. divide by a factor > 1) and repeat the check. If the added perturbation breaks convexity at the sampled points, shrinking β_{k+1} weakens the deformation until a convex interpolant exists again.

Interpolation Check via Dual Linear Program

The interpolation condition requires, for each i, feasibility of

$$U_\beta^{k+1}(z_0^{(j)}) \;\geq\; U_\beta^{k+1}(z_0^{(i)}) + g_i^\top (z_0^{(j)} - z_0^{(i)}) \quad \forall j.$$

By Farkas' Lemma, these inequalities are infeasible for some i if and only if there exist coefficients $\alpha_j \geq 0$ such that

$$\sum_{j=1}^{m} \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0, \qquad \sum_{j=1}^{m} \alpha_j \big(U_\beta^{k+1}(z_0^{(j)}) - U_\beta^{k+1}(z_0^{(i)}) \big) < 0.$$

Since linear programs cannot handle strict inequalities, we solve

$$\min_{\alpha \geq 0} \sum_{j=1}^{m} \alpha_j \big(U_\beta^{k+1}(z_0^{(j)}) - U_\beta^{k+1}(z_0^{(i)}) \big) \quad \text{s.t.} \; \sum_{j=1}^{m} \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0.$$

If this LP is unbounded for any i, the interpolation condition fails and U_β^{k+1} is not convex on the samples. **Note:** Solving the dual LP is computationally efficient when m is large.

Intuition

Given sample locations $z_0^{(i)} \in \mathbb{R}^d$, function values $U(z_0^{(i)}) \in \mathbb{R}$, and candidate subgradients $g_i \in \mathbb{R}^d$, we seek a convex function that interpolates these values.

A canonical convex interpolant is

$$f(z) = \max_{i=1,\ldots,m} \left\{ U(z_0^{(i)}) + g_i^\top (z - z_0^{(i)}) \right\}.$$

Each term $U(z_0^{(i)}) + g_i^\top (z - z_0^{(i)})$ defines an affine supporting hyperplane passing through $(z_0^{(i)}, U(z_0^{(i)}))$.

Taking the pointwise maximum forms the upper envelope of these hyperplanes, producing a convex function that lies above all supports.

Since the maximum of affine (hence convex) functions is convex, the resulting function f is convex by construction.

ICNN Basis Functions: Convexity by Construction

In the second approach, each basis function

$$F_\ell^k : \mathbb{R}^d \to \mathbb{R}$$

is parameterized as an *Input Convex Neural Network* (ICNN). An ICNN is designed so that convexity in the input z is guaranteed *structurally*, not enforced post hoc. Each hidden layer has the form

$$h_{t+1}(z) = \sigma\left(W_{t+1}^{(z)} z + W_{t+1}^{(h)} h_t + b_{t+1}\right),$$

where:

- $W_{t+1}^{(h)} \geq 0$ entrywise,
- σ is convex and non-decreasing (e.g. ReLU, softplus),
- affine terms in z are unrestricted.

Because convexity is preserved under affine maps, nonnegative linear combinations, and composition with convex non-decreasing functions, the resulting function $F_\ell^k(z)$ is convex in z by construction.

Results using ICNNs

Special Case: $\alpha = 1$: $T(x) = x + \sum_{k=1}^{K} \beta_k \nabla F_k(x)$, Identity map plus a convex perturbation from the ICNN terms.

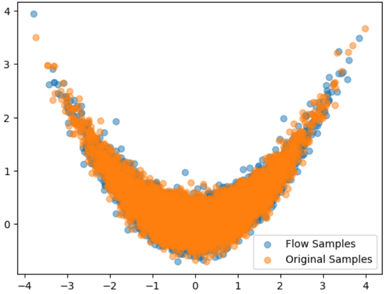

Figure: Banana distribution $p(x,y) \propto \mathcal{N}(x; 0, 1)\, \mathcal{N}\left(y; \frac{x^2}{4}, 0.2\right)$

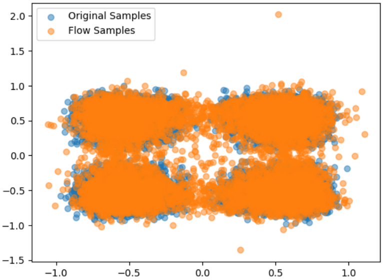

Figure: Convex map $F(x,y) = x^4 + y^4$, visualized by applying ∇F^{-1} to a bivariate normal distribution

Conclusion & Future Work

Conclusion:

- Successfully implemented and debugged a classical non-parametric normalizing flow.
- Developed and tested a modern ICNN-based flow, demonstrating its power and theoretical guarantees.

Future Work:

- Finish implementing both proposed approaches in code.
- Draw clear and effective comparisons between the two approaches.
- Apply these models to higher-dimensional data.

Thank You for Listening!

Questions?

References

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